



BEDOK SOUTH SECONDARY SCHOOL 2026 PRELIMINARY EXAMINATION

4E 5N

CANDIDATE
NAME

CLASS

INDEX
NUMBER

MATHEMATICS

Paper 1

4052/01

26 August 2026

Candidates answer on the Question Booklet.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

	Questions	Marks Deducted
Accuracy		-1 / -2
Presentation		
Units		

For Examiner's Use
90
Student check and sign
Parent/Guardian Signature

2 *Mathematical Formulae*

Compound Interest

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$

Answer **all** the questions.

- 1** By rounding each number to 1 significant figure, find an estimate for the value of

$$\frac{\sqrt{899} \times 0.197}{\pi}.$$

Show the numbers you use.

Answer [1]

- 2 (a)** Express 3528 as a product of its prime factors.

Answer [2]

- (b)** Hence, find the smallest positive integer k such that $3528 \times k$ is a perfect cube.

Answer $k =$ [1]

- 3 A bicycle is sold at a discount of 19%.
After the discount, it costs \$647.

Find the original price of the bicycle. Give your answer correct to the nearest dollar.

Answer \$ [2]

- 4 At a sports event, the ratio of the number of boys to girls in the sports hall is 16 : 11 .
During an intermission, 23 boys joined while 15 girls left the event.
The ratio of the number of boys to girls in the sports hall became 7 : 3 .

Find the number of girls in the sports hall at the end of the event.

Answer [3]

- 5 (a) Solve the inequalities $-66 \leq 12(7 - 9x) < 57$.

Answer [2]

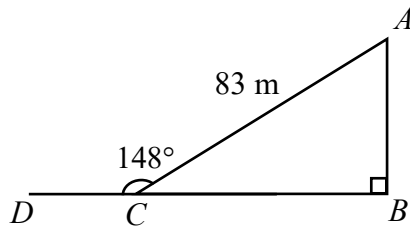
- (b) Expand and simplify $(3a - 2b)(2a + 5b)$.

Answer [2]

6 Simplify $\frac{x^2-9}{2x^2+5x-3} \div \frac{x-9}{2x-1}$.

Answer [3]

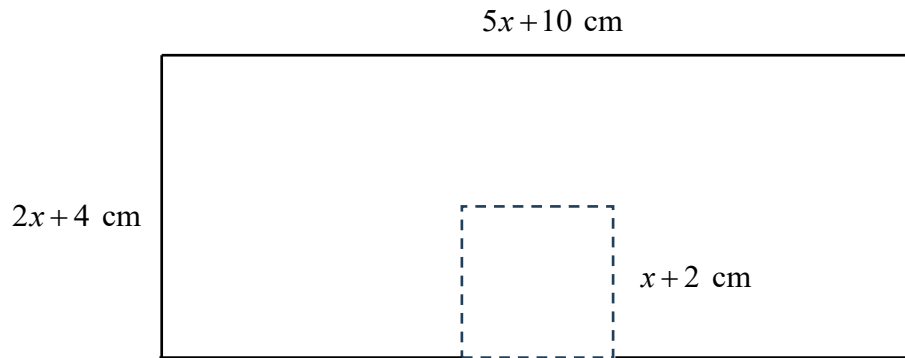
- 7 In the diagram, ABC is a right-angled triangle. D is a point of BC produced such that angle $ACD = 148^\circ$.



Given that $AC = 83 \text{ m}$, calculate the length of AB .

Answer m [2]

- 8 The diagram shows a rectangular sheet of length $5x + 10$ cm and breadth $2x + 4$ cm from which a square of side $x + 2$ cm is cut. All dimensions measured are integer values. The area of the remaining sheet is A cm².



- (a) Show that $A = 9(x + 2)^2$.

Answer

[3]

- (b) Rearrange the formula in **part (a)** to make x the subject.

Answer $x = \dots\dots\dots$ [2]

- (c) Show, with calculations, that A must be at least 9.

Answer

[2]

- 9 An energy company recorded the monthly electricity consumption of a sample of 80 households in the table below.

Monthly Electricity Consumption, x (kWh)	Number of households
$150 \leq x < 250$	9
$250 \leq x < 400$	26
$400 \leq x < 500$	20
$500 \leq x < 700$	18
$700 \leq x < 1000$	7

- (a) Calculate an estimate for
- (i) the mean monthly electricity consumption of the households,

Answer kWh [2]

- (ii) the standard deviation of the monthly electricity consumption.

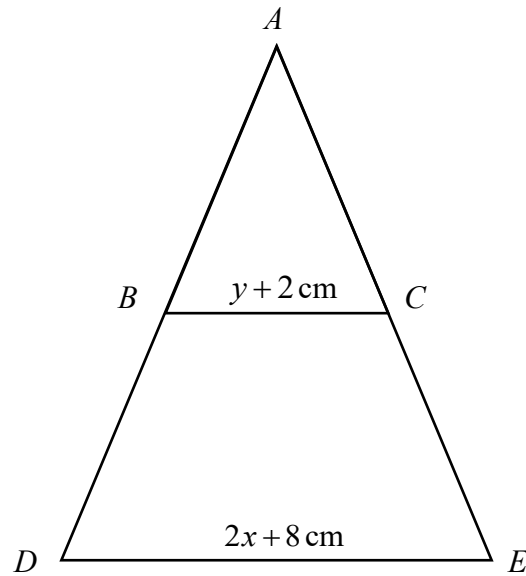
Answer kWh [1]

- (b) Suggest why a bar graph may not be appropriate to represent this data.

.....

..... [1]

- 10 The diagram shows a pair of similar triangles ABC and ADE enlarged by a scale factor of 2. The lengths of a pair of corresponding sides BC and DE are given in the diagram.



- (a) Show that $x = y - 2$.

Answer

- (b) The difference in lengths of the corresponding sides BC and DE is 10 cm.

[2]

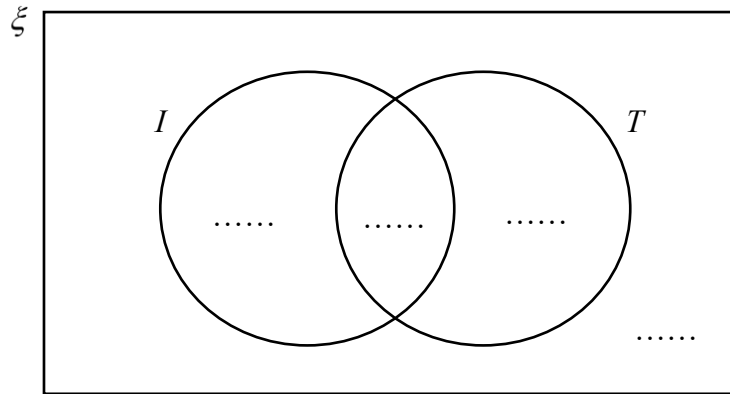
By forming a pair of simultaneous equations, solve for x and y .

Answer $x = \dots\dots\dots$

$y = \dots\dots\dots$ [2]

- 11 In a class of 60 students, the set I represents the set of students with an Instagram account and the set T represents the set of students with a TikTok account.

36 students own an Instagram account and 28 students own a TikTok account.
The number of students that own both an Instagram and a TikTok account is 10.



- (a) In the Venn diagram above, fill in the blanks with the number of students in each of the subsets. [2]
- (b) Find the value of $n[(I \cap T') \cup T]$.

Answer [2]

- (c) State what the set $(I \cap T) \cup (I \cup T)'$ represents.

.....

..... [1]

- 12 (a) Express as a single fraction in its simplest form $\frac{x}{x-2} - \frac{2}{x+1} - \frac{x+4}{(x-2)(x+1)}$.

Answer [3]

- (b) Solve the equation $\frac{2x-1}{3} - \frac{x+4}{2} = \frac{5}{6}$.

Answer $x =$ [2]

- 13** Mr Daniel deposited \$15 000 in a savings account. The account earns compound interest at $r\%$ per annum, compounded half-yearly.

At the end of 3 years, he withdrew all the money in the account and received \$17 107.33.

Find the value of r .

Answer $r = \dots\dots\dots$ [3]

- 14** A student simplified an expression and obtained

$$\frac{16a^5b^{-3}}{D} = 2a^2b$$

Express D in terms of a and b . Leave your answer in positive indices.

Answer $D = \dots\dots\dots$ [2]

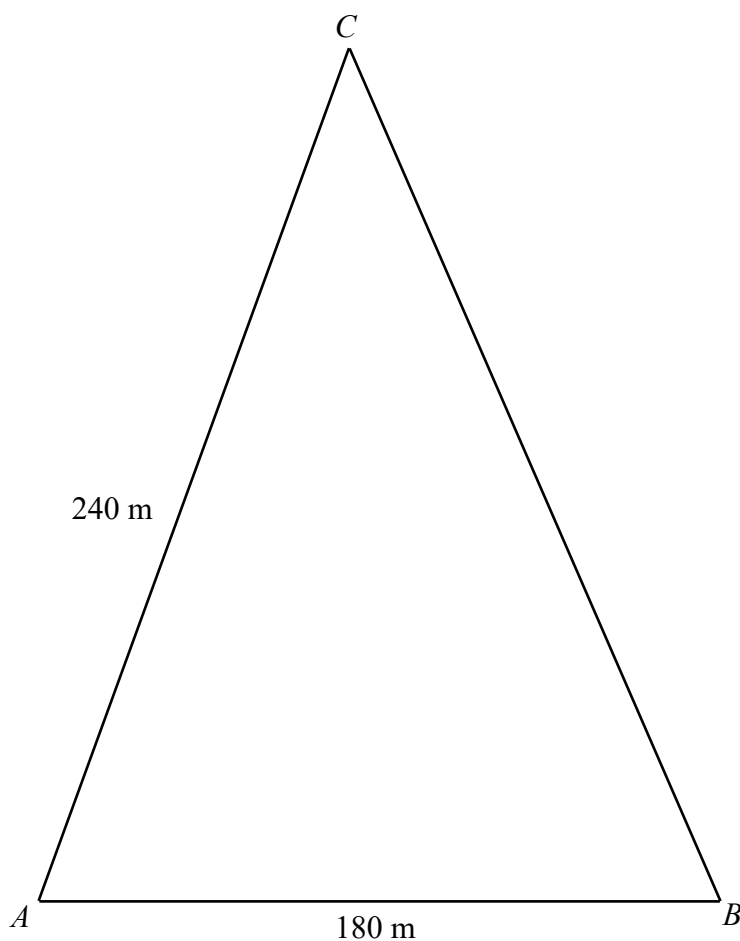
- 15 In triangle ABC , $\overrightarrow{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 0 \\ c \end{pmatrix}$.

If $|\overrightarrow{BC}| = \sqrt{45}$, find the possible values of c .

Answer $c = \dots\dots\dots$ or $\dots\dots\dots$ [4]

- 16** The diagram below shows a triangular piece of land ABC .
 $AB = 180$ m, $AC = 240$ m and $\angle BAC = 70^\circ$.

The diagram is drawn accurately to the scale 1 cm : 20 m.



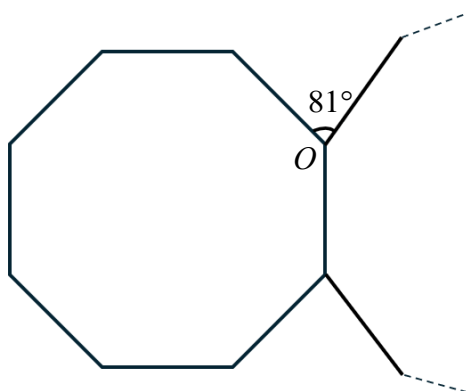
- (a) Construct the perpendicular bisector of AB . [1]
 (b) Construct the angle bisector of angle BAC . [1]

A 50 m tower is built at the intersection of the two bisectors constructed in **part (a)** and **part (b)**.

- (c) Find the angle of elevation of the top of the tower from A .

Answer [3]

- 17 A regular octagon and a regular n -sided polygon is placed beside each other such that they share a common side as shown in the diagram below.



The angle at O , formed between the sides of the polygons, is 81° .

- (a) Determine n .

Answer $n = \dots\dots\dots$ [3]

- (b) Explain, with calculations, why another regular polygon cannot fit exactly at O without gaps.

.....

.....

[2]

- 18** A bag contains a total of 21 red and blue tokens. There are x more red tokens than blue tokens in the bag. Two tokens are drawn, one at a time, with replacement.

- (a) Show that the probability of drawing two blue tokens is $\left(\frac{21-x}{42}\right)^2$.

Answer

[2]

- (b) It is given that the probability of drawing two blue tokens is $\frac{1}{9}$.

Find the number of red tokens in the bag.

Answer [3]

- (c) Find, as a fraction in its simplest form, the probability that at least one of the tokens drawn is red.

Answer [1]

- 19 The coordinates of the vertices of triangle PQR are

$$P(3,7), Q(7,9) \text{ and } R(2,9)$$

Determine, with calculations, if triangle PQR is a right-angled triangle.

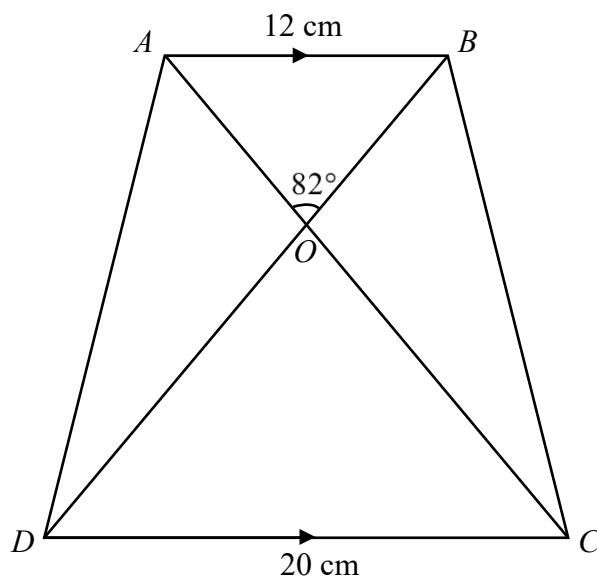
.....
 [3]

- 20 The graph of $x^2 - 8x + k$ touches the line $y = 13$ at exactly one point.

By expressing $x^2 - 8x + k$ in the form $(x - a)^2 + b$, find the value of k .

Answer $k = \dots\dots\dots$ [2]

- 21 $ABCD$ is a trapezium with $AB \parallel DC$. The diagonals AC and BD intersect at O .



- (a) Name a pair of similar triangles and show that they are similar. Give a reason for each statement you make.

Answer

[2]

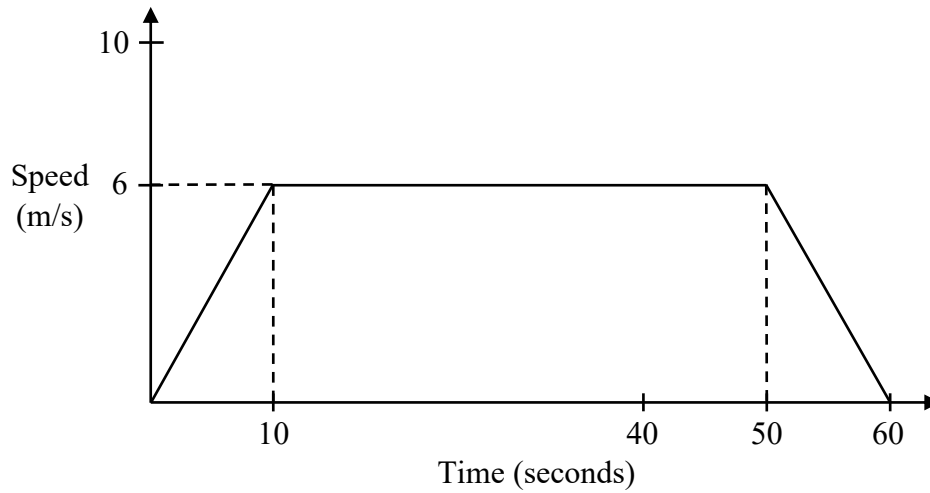
It is given that $AB = 12$ cm, $CD = 20$ cm and angle $AOB = 82^\circ$.

- (b) If $AC = 24$ cm, find angle ABO .

Answer [3]

- 22** Alicia and Bernard begin cycling at the same time. They cycle along the same route in the same direction.

The speed-time graph below shows the motion of Alicia.



- (a)** Calculate the average speed of Alicia during her entire journey.

Answer m/s [2]

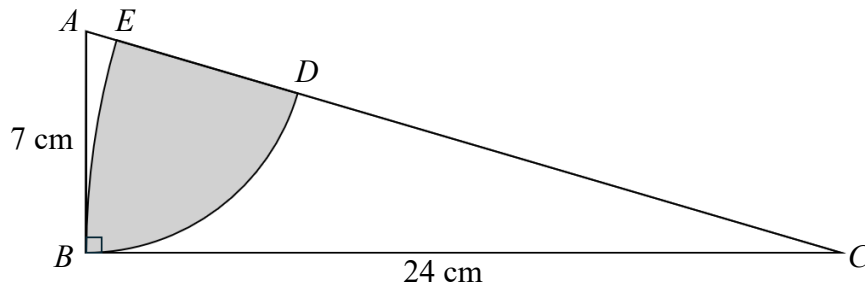
- (b)** Bernard accelerates uniformly from rest to 10 m/s in 40 seconds before riding at that constant speed for the next 20 seconds.

Draw the speed-time graph for Bernard on the same axes above. [1]

- (c)** Determine the time at which Bernard next meets Alicia again during the journey.

Answer $t =$ s [3]

- 23** ABC is a right-angled triangle with $AB = 7$ cm and $BC = 24$ cm. D and E are points on AC such that DAB and ECB are sectors of circles with centre A and C respectively.



- (a)** Show that angle ACB is 0.28379 radians, correct to 5 decimal places.

Answer

[1]

- (b)** Calculate the area of the unshaded region ABE .

Answer cm^2 [2]

- (c)** Find the perimeter of the shaded region.

Answer cm [3]